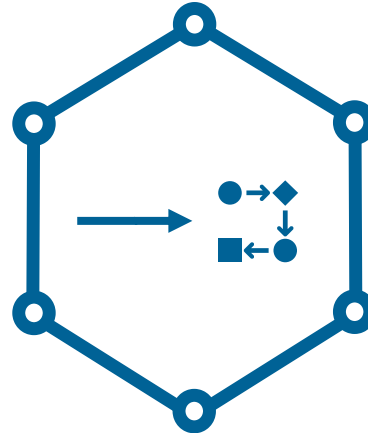




Knowledge Graph Creation Record Linkage

Bonus

Emanuel Sallinger



STANFORD

Slides courtesy of Vinay K. Chaudhri
University of Stanford

Example

	Table A		
	Company	City	State
a ₁	AB Corporation	New York	NY
a ₂	Broadway Associates	Washington	WA
a ₃	Prolific Consulting Inc.	California	CA

	Table B		
	Company	City	State
b ₁	ABC	New York	NY
b ₂	Prolific Consulting	California	CA

a₁=b₁

a₃=b₂

Inexact Inference

In practice, millions of records

Approach

- Blocking Followed by Matching

	Table A		
	Company	City	State
a ₁	AB Corporation	New York	NY
a ₂	Broadway Associates	Washington	WA
a ₃	Prolific Consulting Inc.	California	CA

	Table B		
	Company	City	State
b ₁	ABC	New York	NY
b ₂	Prolific Consulting	California	CA

Blocking

<a₁,b₁>

<a₃,b₂>

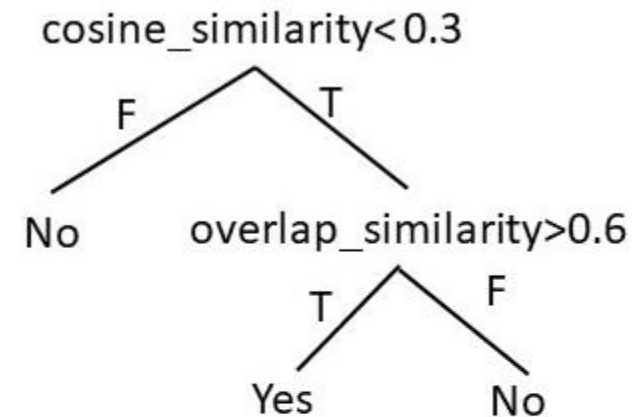
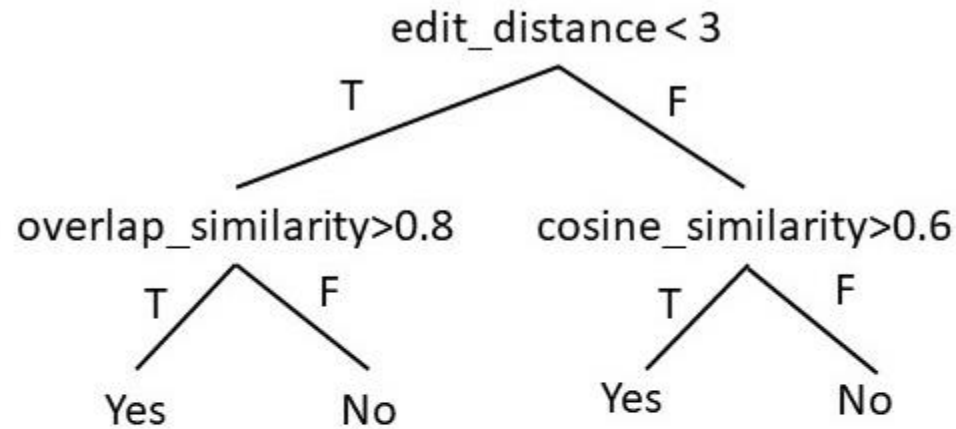
Overview of the algorithm

- Express the blocking/matching rules as a random forest
- Use Active Learning to build the random forest
- Efficient application of rules through indexing

Random Forest

- Consists of a set of set of rules
- Each rule for selects records that use similarity functions
 - Edit distance
 - Overlap similarity
 - Cosine similarity

Random Forest



r_1 : (edit_distance ≥ 3) and (cosine_similarity > 0.6) $\rightarrow$ match

r_2 : (edit_distance < 3) and (overlap_similarity > 0.8) $\rightarrow$ match

r_3 : (cosine_similarity ≥ 0.3) and (overlap_similarity > 0.6) $\rightarrow$ match

Random Forest

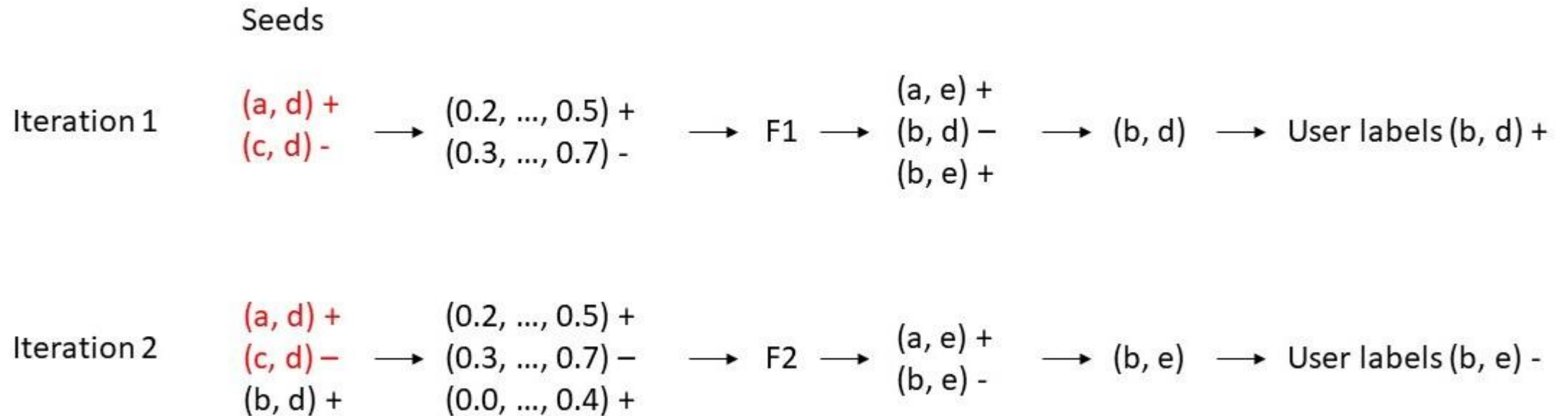
- General principles for selecting similarity functions
 - Numeric-valued attributes such as age, weight, price, etc.
 - exact match, absolute difference, relative difference, and Levenstein distance
 - String-valued attributes
 - edit distance, cosine similarity, Jaccard similarity, and TF/IDF functions.

Active Learning

- Randomly select pairs from the two data sets
 - Ask the users to label them
- Use similarity functions to obtain features
- Learn random forest
- Apply the learned rules to new selected pairs
 - Evaluate the rules
- Iterate

Active Learning

- Source 1: (a,b,c) Source 2: (d,e)



Active Learning

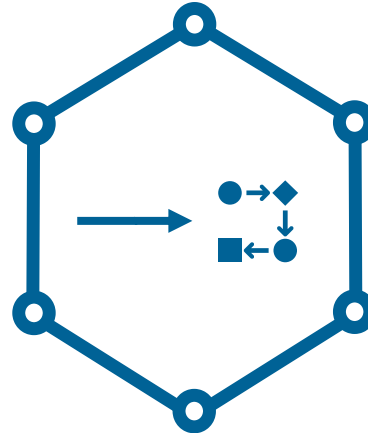
- Once the learning algorithm converges, present the rules to the user
- Retain the rules validated by the user

Rule Application

- Leverage indexing for efficient application of rules
 - Suppose we need to check Jaccard similarity to movie “Sound of Music”
 - If the similarity needs to be greater than 0.7, we need to consider only those movies with length between 3×0.7 , and $3/0.7$, ie, between 2 and 4
 - An index on the length of movies can help us select which movie records to consider

Blocking vs Matching

- Same algorithmic outline is used except
 - The matching rules are more exact/price
 - The matching is usually verified through human intervention



Knowledge Graph Creation Record Linkage

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